

AI-Generated Content (Deepfakes)

CompTIA SecAI+: AI Security Certification Complete Exam Prep 2026 | Section 19: AI

1. Beyond the Video Call: The Full Synthetic Media Threat Taxonomy

The term 'deepfake' entered public awareness through manipulated celebrity videos, but the threat taxonomy is far broader and more operationally relevant than that cultural starting point suggests. Synthetic media encompasses four distinct asset classes, each with its own attack surface and countermeasure set.

Asset Class	Primary Generation Method	Attack Scenarios	Lead Countermeasure
Video deepfakes	GAN face-swap / diffusion real-time	CFO fraud calls, executive impersonation meetings	Out-of-band verification + C2PA
Voice clones	Neural TTS trained on 1-60 sec audio	CEO fraud calls, vishing, account takeover	Shared code words + FIDO2 MFA
Image fakes	StyleGAN3, diffusion (DALL-E, Midjourney)	Synthetic identity fraud, fake KYC, fake vendor contacts	C2PA provenance metadata
Text generation	LLMs (GPT-4, Claude, Gemini)	Style-mimicry BEC, AI-written phishing, fake review campaigns	DMARC/DKIM enforcement + behavioral email analysis

Understanding this taxonomy prevents the mistake of building a deepfake defense that addresses only video, while voice cloning and image fraud proceed unchecked. Each asset class requires its own detection approach and its own process control, though out-of-band verification provides a unifying procedural safeguard across all four.

2. The GAN Architecture: Why Deepfakes Keep Getting Better

Generative Adversarial Networks are the foundational architecture behind most video deepfake systems. Understanding how they work explains why detection is an arms race rather than a solved problem.

A GAN consists of two neural networks in competition. The generator creates synthetic output -- in video deepfakes, replacing a face in a video frame with a synthesized version trained on the target's reference images. The discriminator's job is to distinguish generated frames from real ones. After thousands of training iterations, the generator has learned to produce output that consistently fools the discriminator -- which means it also fools human visual systems trained to the same fidelity level.

Diffusion models (used in Stable Diffusion, Midjourney, DALL-E 3) take a different approach: starting from random noise and iteratively denoising toward a target image. They produce comparable or superior image quality without the adversarial training loop, and the absence of GAN-specific artifacts makes them harder to detect with first-generation detection tools.

- Reference data required for video deepfake: 300-500 images, easily scraped from LinkedIn and press releases.

- Voice clone threshold: ElevenLabs achieves recognizable clone from 60 seconds; Microsoft VALL-E demonstrated 3 seconds is sufficient for recognizable output.
- Hardware threshold (2024): consumer-grade NVIDIA RTX 4090 runs real-time face-swap -- no GPU cluster required.
- Detection lag: new generation architectures regularly outpace detection models trained on previous-generation artifacts.

3. Voice Cloning: The Lowest-Barrier Entry Point

If video deepfakes require meaningful technical setup, voice cloning is nearly frictionless and represents the attack vector most likely to be encountered in enterprise environments today. The attack surface for executive voice data is enormous and largely unmanaged.

Real-World Incident -- UK Energy Firm CEO Fraud (2019): Attackers cloned the voice of a German parent company CEO using neural TTS technology and called the UK subsidiary managing director. The managing director transferred 220,000 euros to an attacker-controlled account. This incident predated current-generation tools; modern voice cloning is substantially more accurate and requires less source material. This case established the attack pattern that 2023-2024 campaigns scaled to video deepfakes.

The attack surface for voice data is poorly understood by most organizations. An executive who records a quarterly video update, appears in a podcast, gives a TED talk, or hosts an earnings call has provided an attacker with all the source material needed. Security teams should treat executive audio as sensitive data -- subject to the same consideration as credentials and PII -- when developing data classification policies.

Voice cloning platforms available in 2024 include ElevenLabs, Resemble AI, PlayHT, and Descript Overdub. All are commercially accessible. High-value OSINT sources for voice data: earnings call recordings on investor relations pages, YouTube interview archives, LinkedIn voice notes, and conference talk recordings.

4. Image Deepfakes and Synthetic Identity Fraud

Static image deepfakes generate a specific and rapidly growing class of attack: synthetic identity fraud. Tools such as StyleGAN3 produce photorealistic faces of people who do not exist -- these images yield no results in reverse image search, because the depicted person has no real-world presence to match against.

Attackers use AI-generated faces to fabricate:

- Employee profiles on LinkedIn and corporate directories -- used for social engineering and to establish fake insider personas.
- Vendor and supplier contacts -- used in supply chain impersonation attacks where the fake contact initiates fraudulent invoicing.
- KYC (Know Your Customer) identity documents -- submitted to financial institutions and cryptocurrency exchanges for fraudulent account establishment.

- Fake security researchers and technical experts -- used to approach genuine professionals with social engineering pretexts, as documented in state-sponsored campaigns targeting defense contractors.

In 2024, multiple cryptocurrency exchanges reported synthetic identity fraud attempts using AI-generated faces on fake passports submitted through KYC verification workflows. The C2PA standard addresses the provenance gap, but KYC workflows accepting document photographs do not yet widely verify C2PA metadata -- this remains an open gap in identity verification infrastructure.

5. Framework Placement: NIST CSF 2.0 and MITRE ATLAS v2.1

Placing deepfake threats within established frameworks is an exam requirement and a practical necessity for building a defensible program.

Framework Element	Deepfake Application	Practical Implication
GV.RM (Risk Management)	Synthetic media must be added to the organizational risk register -- most registers do not yet include it as a named category	Add deepfake risk with defined likelihood and impact ratings; assign risk owner
DE.AE (Adverse Event Analysis)	Detection of anomalous communications -- video calls and voice interactions diverging from known biometric baselines	Integrate out-of-band callback procedures into DE detection workflows for high-stakes requests
RS.CO (Response Communications)	Post-incident communications must themselves be authenticated to avoid confusion with further synthetic media	Pre-establish cryptographically signed communication channels for incident response use
MITRE ATLAS v2.1 -- Craft Adversarial Data	Attacker trains or fine-tunes a model specifically to deceive a human target using synthetic output	Deepfake attacks differ from traditional social engineering: they involve technical manipulation of model outputs, not just persuasion

6. Breaking the Attacker's Kill Chain

From a red team perspective, a deepfake campaign follows a predictable sequence. Defenders who understand this sequence can identify which interventions offer the highest leverage.

- Phase 1 -- OSINT Gathering: Attacker collects reference material from LinkedIn, press releases, earnings calls, and company websites. Intervention: minimize publicly available executive biometric data and audit what is genuinely necessary to publish.
- Phase 2 -- Target Selection: Attacker identifies the victim and the scenario. Intervention: limit visibility of org chart detail and reporting relationships in public directories.
- Phase 3 -- Synthesis: Attacker generates the synthetic media using commercially available tools. Intervention: register executive biometric baselines with internal detection systems; deploy endpoint tools that flag virtual camera drivers.

- Phase 4 -- Delivery: Attacker delivers content through a plausible channel -- spoofed meeting invite, forwarded video message, voice call. Intervention: enforce DMARC at p=reject; deploy display name alerting for external emails claiming to be internal executives.
- Phase 5 -- Exploitation: Attacker makes the fraudulent request. Intervention: mandatory out-of-band verification for all financial transfers, credential changes, and sensitive access provisioning -- regardless of how convincing the communication appeared.

7. Detection Technology: Capabilities and Limits

Tool / Approach	Detection Method	Key Limitation
Intel FakeCatcher	rPPG signal analysis -- blood flow patterns visible in pixel color changes that synthesized faces fail to replicate consistently	Fails against diffusion-generated video not involving face-swap over real background video
Microsoft Video Authenticator	GAN fingerprint detection in pixel noise; temporal audio-video correlation analysis	Accuracy degrades against models specifically trained to suppress GAN artifacts
Sensity AI / Reality Defender	Multi-model ensemble detection; API-accessible for real-time screening	False positive and negative rates remain meaningful; not suitable as standalone verdict system
rPPG Physiological Analysis	Human faces show blood-flow-driven color oscillation; synthesized faces often fail to replicate it	Requires controlled lighting and adequate video resolution to detect signal reliably
Behavioral Anomaly Detection	Flags video calls with unnatural blink rates, lip-sync delays, or lighting inconsistencies	High-quality real-time deepfakes have reduced many of these artifacts to below reliable human detection threshold

Exam-critical point: detection tools are most valuable as triage mechanisms flagging content for human verification -- not as definitive verdict systems. A control architecture that depends on detection succeeding will fail when generation quality exceeds detection capability.

8. Content Provenance: C2PA as the Structural Defense

Structural Insight: The most durable countermeasure is provenance -- knowing where content came from and whether it has been tampered with -- not detection of whether content appears fake. C2PA (Coalition for Content Provenance and Authenticity) implements this principle at the standards level.

C2PA v2.0 (released 2024) defines how cameras, smartphones, and editing software embed cryptographically signed manifests directly into image, video, and audio files. These manifests travel with the content and record the creation device, creation timestamp, and editing history. Verifying a manifest against a certificate authority confirms provenance and chain of custody.

C2PA adopters as of 2024 include Adobe (Photoshop, Premiere), Leica, Nikon, Sony, and Microsoft. The ContentCredentials.org initiative provides a public verification interface.

- Valid, unbroken C2PA manifest: high confidence in provenance -- the content originated from a registered device and has a documented editing history.
- Absent or broken manifest: not automatically fake -- many legitimate recordings lack C2PA support. In a high-stakes context, absence of provenance should trigger out-of-band verification.
- Organizational implication: mandate C2PA-compliant recording equipment for executive video communications intended for distribution.

Memory anchor: C2PA is to video and images what DKIM is to email -- a cryptographic signature that lets the recipient verify origin without trusting the delivery channel. Just as DKIM alone does not stop all email fraud, C2PA alone does not stop all synthetic media -- but it closes the easiest attack vectors and enables structured verification.

9. Verification Protocols: The Process That Beats the Technology

The organizational defense that most consistently prevents deepfake fraud succeeds regardless of how technically sophisticated the deepfake is: mandatory out-of-band verification for high-stakes decisions.

- Channel independence: the verification channel must differ from the channel through which the request arrived. Video call request -> phone verification to pre-registered number. Phone request -> authenticated messaging app or hardware token.
- Pre-established contact information: the verification number must come from a directory established before any incident -- never from a number provided by the requester.
- Mandatory applicability regardless of convincingness: the protocol applies even when the communication appears entirely legitimate. The moment an employee thinks 'this seems real so I do not need to verify,' the control has failed by design.
- Shared rotating code words: established each quarter, these short phrases are highly effective because a cloned voice cannot produce the correct word in context if the attacker does not know it.

Exam application: when a scenario describes a deepfake fraud incident and asks what control would have prevented it, out-of-band verification is the correct answer -- not deepfake detection software. Process controls that are consistently followed do not depend on technology accuracy.

10. Employee Training That Changes Behavior

Security awareness training fails when it only shows employees examples of visually detectable fakes. Current deepfakes are often not visually detectable. Effective training does three things differently:

- Shows near-perfect fakes: demonstrate that 'look harder at video quality' is not the goal. The cognitive tripwire that used to catch impersonation -- 'something seems off' -- no longer fires reliably against high-quality synthetic media.
- Reframes the behavior target: the key behavior is 'always verify through a second channel before acting on a video or voice request,' not 'identify whether this video looks fake.' Employees can reliably do the former; the latter requires technology they do not have.
- Addresses the liar's dividend: students must understand that the existence of deepfakes lets real evidence be dismissed as fake. Attackers exploit this in both directions -- making false content believed, and making real evidence doubted. Employees who understand this dynamic are harder to manipulate in either direction.

Vendors including KnowBe4 and Proofpoint now offer simulated deepfake attack modules -- the direct analogue of phishing simulations. Simulation exercises consistently close the gap between awareness training and behavioral change.

11. Legal and Regulatory Landscape

Jurisdiction / Instrument	Scope and Requirement	Practitioner Implication
California AB 602 / AB 730	Deepfake pornography (civil right of action) and political deepfakes within 60 days of election (criminal liability)	Organizational policy should prohibit generation of deepfakes depicting real individuals without consent
US DEEPFAKES Accountability Act (proposed)	Mandatory disclosure requirement for AI-generated media; criminal penalties for malicious deployment	Incident response plans must include legal notification steps when deepfake fraud is confirmed
EU AI Act (2024), Article 52	AI-generated content that could be mistaken for real must be labeled; covers synthetic audio, video, and text in commercial contexts	Deploying unlabeled synthetic media in commercial contexts creates legal exposure under GDPR jurisdiction
C2PA / ContentCredentials.org	Industry standard for content provenance -- not yet legally mandatory but referenced in regulatory guidance	Adopting C2PA for executive communications provides both technical and compliance posture benefits

Key Terms Glossary

Term	Definition
GAN (Generative Adversarial Network)	A machine learning architecture using two competing networks - a generator and a discriminator -- to produce synthetic output that progressively fools detection. The foundational architecture for most video deepfake systems.
Deepfake	Synthetic media (video, audio, or image) in which a person's likeness is replaced or fabricated using generative AI. Encompasses video face-swap, voice cloning, and AI-generated still images of real people.
Voice Cloning	Generation of synthetic speech replicating a specific person's voice characteristics using a neural TTS model trained on reference audio. Commercially available in 2024 from as little as 60 seconds of source material.
rPPG (Remote Photoplethysmography)	A technique for detecting blood flow patterns in human faces by analyzing pixel color changes at the capillary level. Real faces show consistent rPPG signals; synthesized faces often do not replicate them accurately.
C2PA (Coalition for Content Provenance and Authenticity)	An industry standard for embedding cryptographically signed manifests in media files, enabling provenance verification.

	Backed by Adobe, Microsoft, Nikon, Sony, and others; v2.0 released 2024.
Liar's Dividend	The phenomenon where the existence of deepfakes enables real evidence to be dismissed as fake, creating strategic ambiguity that adversaries deliberately exploit in both offensive and defensive directions.
Diffusion Model	A generative AI architecture producing images by iteratively denoising random noise toward a target distribution. Used in Stable Diffusion, Midjourney, and DALL-E 3; produces fewer GAN-specific artifacts, complicating detection.
StyleGAN3	NVIDIA's generative model for producing photorealistic human faces with no real-world counterpart. Output yields no results in reverse image search, enabling synthetic identity fraud.
MITRE ATLAS v2.1	A framework cataloguing adversarial tactics targeting machine learning systems. Categorizes deepfake generation under the 'Craft Adversarial Data' tactic.
Out-of-Band Verification	Confirming the authenticity of a request through a communication channel different from the one through which the request was received, using contact information pre-established before the incident.

Quick Revision Checklist

- Deepfakes span four asset classes: video, voice, image, and text -- each with distinct tools, attack scenarios, and countermeasures.
- GAN architecture: generator vs. discriminator competition drives quality improvement; diffusion models produce fewer detectable artifacts.
- Voice cloning requires minimal source material (seconds to minutes); executive audio is an unmanaged attack surface in most organizations.
- Synthetic identity fraud uses StyleGAN-generated faces that yield no reverse image search results -- defeating a common verification shortcut.
- NIST CSF 2.0 mapping: GV.RM (risk register), DE.AE (detection), RS.CO (authenticated comms). MITRE ATLAS: Craft Adversarial Data.
- Attacker kill chain: OSINT -> profiling -> synthesis -> delivery -> exploitation. Highest-leverage break points: OSINT reduction and out-of-band verification.
- Detection tools (FakeCatcher, Video Authenticator, Sensity AI) are triage aids, not definitive verdict systems. Detection is an arms race.
- C2PA provides cryptographic provenance -- the analogue of DKIM for media files. Content without a valid manifest should trigger verification in high-stakes contexts.
- Out-of-band verification is the exam answer for 'what would have prevented this?' -- not detection technology.
- Liar's dividend: deepfakes create ambiguity in both directions -- false content is believed, and real evidence is doubted.

- EU AI Act Article 52: AI-generated content that could be mistaken for real must be labeled. Unlabeled synthetic media of real people creates legal exposure.
- Employee training goal: change behavior (always verify via second channel) -- not improve visual detection ability.

10 Exam-Based MCQs -- AI-Generated Content and Deepfakes

Test yourself after reading. These questions are written in real exam style -- scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. Scenario: A finance department employee at a multinational receives a video call from what appears to be the CFO and two colleagues, instructing an urgent wire transfer of 2.4 million USD to a new vendor account. The call lasts twelve minutes and is visually convincing. The employee transfers the funds. Forensic analysis confirms every participant on the call was a deepfake generated from publicly available executive video recordings. What control would MOST effectively have prevented the loss?

- A) Deploy deepfake detection software on all video conferencing endpoints
- B) Require out-of-band verification through a pre-registered contact for all wire transfers above the defined threshold
- C) Implement DMARC at p=reject to prevent domain spoofing in the meeting invitation
- D) Train employees to identify visual artifacts in synthetic video

Answer: B **Explanation:** B is correct because out-of-band verification through a pre-established contact would have required the attacker to compromise a second, independent channel -- which the deepfake attack cannot achieve. A is wrong because detection software would have to succeed in real-time against a high-quality deepfake; current detection tools cannot guarantee this, and failure would not have changed the outcome. C is wrong because DMARC prevents email domain spoofing -- the attack vector here is a video call, not an email, so DMARC would not have stopped the deepfake call itself. D is wrong because training employees to spot visual artifacts is ineffective against high-quality deepfakes -- the Hong Kong case showed the call was visually convincing to trained observers.

Q2. Scenario: A CISO is presenting a board-level risk briefing following a deepfake incident at a competitor. The board asks which CSF 2.0 function requires the organization to formally recognize and treat synthetic media as an organizational risk requiring a named entry in the risk register. Which NIST CSF 2.0 function is MOST directly responsible for ensuring deepfake risk is formally identified and treated at the organizational level?

- A) DE.AE (Adverse Event Analysis)
- B) PR.AC (Access Control)
- C) GV.RM (Risk Management)
- D) RS.CO (Response Communications)

Answer: C **Explanation:** C is correct because GV.RM covers risk identification and management, including adding synthetic media as a named organizational risk category in the risk register -- the structural starting point for any deepfake program. A is wrong because DE.AE covers detection of anomalous events -- relevant to detecting deepfake incidents in progress, but not to formal program establishment at the

governance level. B is wrong because PR.AC covers access control -- relevant to protecting sensitive systems, but not to managing deepfake risk as an organizational category. D is wrong because RS.CO covers communications during incident response -- important for authenticated post-incident messaging, but this is a response function, not a risk governance function.

Q3. The security team must recommend either deepfake detection software or a mandatory out-of-band verification protocol as the PRIMARY control for high-value financial approvals. Which argument BEST justifies the verification protocol?

- A) Detection software is too expensive and difficult to deploy at enterprise scale
- B) The verification protocol remains effective even when the deepfake is undetectable by current technology
- C) Detection software cannot be integrated with existing video conferencing platforms
- D) The verification protocol eliminates the need for any technical controls

Answer: B **Explanation:** B is correct because it identifies the fundamental security argument: a verification step that is consistently followed provides protection regardless of detection accuracy. Even a perfect deepfake that defeats all detection tools cannot satisfy the verification requirement unless the attacker also compromises the pre-registered out-of-band channel. A is wrong because cost and deployment complexity are practical concerns, not security reasoning -- a detection system that actually worked would still justify its cost. C is wrong because integration capability is an implementation detail, not a security argument -- and it is also not accurate that detection tools cannot be integrated with video platforms. D is wrong because the verification protocol complements technical controls; it does not eliminate the need for them in a defense-in-depth architecture.

Q4. A GAN-based deepfake detection tool achieves 94% accuracy in lab testing against current deepfakes. Which limitation is MOST important to communicate to leadership considering this as the primary deepfake defense?

- A) The tool works only on video content, not voice or static image deepfakes
- B) Detection accuracy degrades as generation quality improves, creating an ongoing arms race dynamic
- C) The tool requires GPU hardware that most organizations do not have available
- D) 94% accuracy will produce unacceptable false positive rates against legitimate executive video

Answer: B **Explanation:** B is correct because the arms race dynamic is the fundamental strategic limitation: detection models are trained on artifacts from current-generation tools, and each new generation of synthesis models produces different artifacts. A tool achieving 94% against today's deepfakes may achieve substantially lower accuracy against next-generation synthesis. A is wrong because scope limitations are real but secondary -- the arms race dynamic affects all detection modalities regardless of asset type. C is wrong because modern deepfake detection tools are typically cloud-accessible or run on standard server hardware -- GPU requirements are not the defining limitation. D is wrong because false positive rate, while a valid concern, is not the most strategically important limitation to communicate; the degradation dynamic over time is more consequential for a primary control decision.

Q5. Which statement BEST distinguishes MITRE ATLAS v2.1's classification of deepfake attacks from its classification of traditional social engineering attacks?

- A) ATLAS classifies deepfakes under Impact tactics; traditional social engineering is outside ATLAS scope
- B) ATLAS classifies deepfakes under Craft Adversarial Data, because the attack involves generating adversarial inputs to a human cognitive system; traditional social engineering does not involve generating adversarial model inputs
- C) ATLAS classifies deepfakes under Exfiltration because the attack extracts behavioral data from the victim
- D) ATLAS does not cover social engineering; it covers only technical exploitation of ML systems

Answer: B **Explanation:** B is correct because MITRE ATLAS v2.1 categorizes deepfake generation under 'Craft Adversarial Data' -- the attacker deliberately generates synthetic media that the human visual/auditory cognitive system processes as authentic. This is technically distinct from traditional social engineering, which manipulates human psychology without crafting adversarial inputs to a model. A is wrong because while deepfakes cause impact, the ATLAS tactic for generation is Craft Adversarial Data, not an Impact tactic. C is wrong because the Exfiltration framing applies to data theft scenarios -- the defining characteristic of deepfake attacks is crafting adversarial input, not extracting information. D is wrong because ATLAS explicitly covers social engineering vectors that target or exploit ML systems.

Q6. A valid, unbroken C2PA manifest attached to a CEO video message confirms which of the following?

- A) The video has not been accessed or copied by unauthorized parties since its creation
- B) The video was generated by a C2PA-compliant device or software and has not been modified outside of recorded editing steps
- C) The video cannot be used as training data for a deepfake model
- D) The video has passed human editorial review and is authorized for distribution

Answer: B **Explanation:** B is correct because a valid C2PA manifest confirms the provenance chain: the content was created by a registered device or application and the editing history recorded in the manifest is cryptographically intact. It does not make claims about access control, downstream use, or human review. A is wrong because C2PA records creation and editing provenance -- it does not track who has accessed or copied the content; that is a DLP and access control function. C is wrong because C2PA provenance does not prevent extraction of frames or audio for use as training data -- it enables provenance verification, not downstream use restriction. D is wrong because C2PA is a technical provenance standard -- it does not include a human review or authorization workflow component.

Q7. Voice biometric authentication systems used in call center IVR are vulnerable to which specific deepfake capability?

- A) Diffusion model image generation producing synthetic identity documents
- B) Neural text-to-speech voice cloning replicating a target's voice characteristics from short audio samples

- C) GAN-based real-time video face-swapping streamed through a virtual camera driver
- D) LLM-generated scripts mimicking the vocabulary patterns of the account holder

Answer: B **Explanation:** B is correct because researchers demonstrated in 2023 that commercial speaker verification systems (voice biometric authentication used in IVR systems) could be defeated by voice clones generated from neural TTS tools using short audio samples. This is the specific vulnerability class relevant to call center account takeover. A is wrong because image generation (diffusion models) affects static images -- it does not affect audio-based authentication systems. C is wrong because video face-swapping operates in the visual domain -- IVR authentication is audio-only and would not be affected by a video deepfake. D is wrong because LLM-generated scripts might help an attacker pass knowledge-based questions, but the specific vulnerability to voice biometric systems is the synthesized voice itself, not the content of what is said.

Q8. What is the liar's dividend and why does it require explicit coverage in deepfake security awareness training?

- A) The financial gain attackers receive from deepfake fraud -- relevant because it explains attacker motivation
- B) The operational friction created when employees distrust all video communications following deepfake awareness training
- C) The phenomenon where the existence of deepfakes enables real evidence to be dismissed as fake, which attackers exploit in both directions -- making false content believed and making real evidence doubted
- D) The legal protection available to organizations that train employees on deepfake risks, limiting liability in fraud incidents

Answer: C **Explanation:** C is correct because the liar's dividend describes the strategic ambiguity created when audiences know deepfakes exist: real evidence can be challenged as possibly synthetic. Attackers exploit this in both directions -- using synthetic media to deceive, and using the concept of deepfakes to create doubt about authentic evidence. Training must address both directions. A is wrong because 'liar's dividend' is not a financial term for attacker profit -- it is a sociological concept describing the asymmetric benefit that accrues to those spreading disinformation when synthetic media undermines general trust in audio-visual evidence. B is wrong because operational over-distrust is a secondary training design concern, not the definition of the liar's dividend. D is wrong because the liar's dividend has no legal liability dimension -- it describes epistemic and social effects on evidence evaluation.

Q9. An organization wants to reduce the OSINT attack surface that could enable a deepfake campaign targeting the CTO. Which action provides the MOST direct reduction in deepfake-relevant attack surface?

- A) Remove the CTO's name and title from the organization's public website and LinkedIn
- B) Review and limit publicly available audio and video recordings featuring the CTO, removing non-essential recordings from investor relations pages and social media
- C) Require the CTO to use a pseudonym in all external written communications
- D) Encrypt all internal video calls end-to-end to prevent interception of live sessions

Answer: B **Explanation:** B is correct because voice and video recordings are the primary training material for voice cloning and video deepfake generation targeting a specific individual. Reducing volume and availability of high-quality audio/video -- particularly from investor relations pages and YouTube hosting long, high-quality recordings -- directly reduces the quality of deepfake an attacker can produce. A is wrong because removing a name removes a text identifier, not biometric data -- attackers targeting a known individual already have their identity; the relevant attack surface is the biometric data, not the name. C is wrong because a pseudonym would not prevent a deepfake attack against a known individual -- voice and facial characteristics are unchanged by using a different name. D is wrong because end-to-end encryption protects call content in transit -- it does not prevent an attacker who has gathered OSINT from public sources from synthesizing training material from that already-collected data.

Q10. Under EU AI Act Article 52, what transparency obligation applies to an organization deploying AI-generated video of a real executive to communicate internally?

- A) The organization must obtain regulatory approval from the EU AI Office before generating any synthetic media depicting real individuals
- B) The organization must label the AI-generated content as synthetic in a manner that is detectable by the audience
- C) The organization must encrypt all AI-generated content with a government-registered key before distribution
- D) The organization must retain all AI-generated content for seven years for regulatory audit

Answer: B **Explanation:** B is correct because Article 52 of the EU AI Act establishes transparency obligations for AI-generated content that could be mistaken for authentic human content, requiring it to be labeled as AI-generated in a detectable manner. This applies to synthetic audio, video, and text generated by AI systems in commercial contexts. A is wrong because the EU AI Act does not create a licensing or prior-approval regime for synthetic media generation -- it creates disclosure and (for high-risk systems) conformity assessment obligations. C is wrong because government-registered key encryption is not a requirement of the EU AI Act -- this is not an element of the regulation. D is wrong because seven-year retention obligations are associated with financial and medical record-keeping requirements under other instruments, not with AI-generated content under the AI Act.